

IntroToArduino.odt

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Introduction to Using the Arduino Development System

ref1: <https://docs.arduino.cc/hardware/uno-rev3/>

IDEA: Make available project boards for the more complex examples.

Syllabus

0. Setting up the Arduino IDE

(see <https://www.arduino.cc/> > For Makers > Go to Documentation > Software > Arduino IDE 2)

1. Blink; the hardware “Hello World” application.

2. LEDs – Some electronics theory so that parts are not damaged.

3. Adding external LEDs to the Arduino aboard. Apply information from #2.

Experiments with six LEDs:

3a. Turn each LED ON; one at a time, then turn each LED OFF, one at a time.

3a1. Straight line code.

3a2. Using an **Array** to control which LED is Blinked.

3a3. Using an Array of **Data Structures** {LED#, State} and one function.

Use the Array as a circular pattern generator.

Use the Array as a scrolling pattern generator; Chase, Ping-pong, Morse Code.

3a4. Using the random function to randomly pick an LED from the Array to turn ON/OFF or Toggle.

Use an Array to record the **State** of an LED to allow toggling it on and off.

3b. Use a **State Machine** to cycle through different display types.

3b1. Use time checks to set the run-time for each state.

3b2. Use a **Table** to control the state sequence.

3c. (Optional) Differential LED drive (Charlieplex). Control multiple LEDs with minimal I/O pins.

3c1. Control six LEDs with three lines.

3c2. Control twelve LEDs with four lines.

3c3. Control twenty LEDs with five lines.

4. 7 Segment LEDs

4a. Single 7-Segment LED

4a1. Using Arduino functions

4a2. Using direct hardware control registers

4b. Multiple 7-Segment LED

4b1. Using direct hardware control registers

4b2. (Optional) Add a one second tic generator to make a clock timer.

5. Reading Buttons and Keypads

6. DC Motor Control

6.1 Simple Transistor Drive

6.2 H-Bridge Drive

6.3 H-Bridge with Quadrature Feedback